Name:______________________________________

Recitation:______________________________________

Math 340
Final Exam
December 17, 2025

Problem	Score	Problem	Score
1		8	
2		9	
3		10	
4		11	
5		12	
6		13	
7		14	
		Total	

All problems are worth 10 points. The exam is closed book, but you may use a calculator and one 8 ½ × 11" sheet of handwritten notes (both sides). You must show your work to receive full credit. Write solutions in explicit form if possible. All problems have a solution that can be found using the techniques of this class.

Pledge:
On my honor, as a student, I have neither given nor received unauthorized aid on this

examination: ____ ____________

(signature) (date)

Name:____________________________________

Problem 1

Find the general solution to $\frac{dy}{dx} = (2x - 1)(y + 1)$

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Problem 2

Find the general solution to $y'' + 7y' + 12y = 17cos(x)$.

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Problem 3

Solve the initial value problem

$$\frac{dy}{dx} = y - xy^2, y(0) = 1.$$

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Problem 4

Solve the initial value problem $x'' + 6x' + 25x = \delta(t)$, $x(0)=0$, $x'(0)=0$.

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Problem 5

Solve the initial value problem. Note that your answer will have an a in it.

$$x'' + 10x' + 16x = 0, x(0) = a, x'(0) = 0$$

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Problem 6

A mass of 5kg is attached to a spring, causing it to stretch 0.1 meters (10 cm). What is critical damping for this spring-mass system. Use g = 9.8 m/sec.

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Problem 7

Find and classify the equilibria for $\frac{dx}{dt} = x(x^2 - 9)(5 - x)$.

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Problem 8

Solve the initial value problem

$$\frac{dx}{dt} = x - y. \qquad x(0) = 0$$

$$\frac{dy}{dt} = 2x + 4y. \quad y(0) = 3$$

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Problem 9

Solve the boundary value problem:

$$x'' + 2x' - 15x = 0 \text{ with } x(0)=2 \text{ and } \lim_{t\to\infty} x(t) = 0.$$

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Problem 10

A comet has perihelion at 0.4 AU and aphelion at 5 AU. What is the period of the comet (in years).

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Problem 11

What is the a priori lower bound for the radius of convergence of the series solution about $x\_0 = 0$ for

$$(x^2 + 4)y'' + (x - 1)y' + 2y = 0, y(0) = 1, y'(0) = 2.$$

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Problem 12

Find the general solution to

$$x^2y'' + 7xy' + 7y = 0.$$

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Problem 13

A two-tank CSTR consists of two 2000 liter tanks. Solution flows into the first tank, from the first tank to the second tank, and out of the second tank all at a rate of 20 liters/min. Suppose 500 grams of a contaminant is introduced to the first tank. What is the maximum concentration reached for the contaminant in the second tank?

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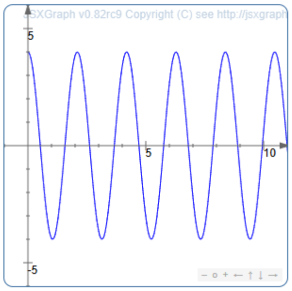
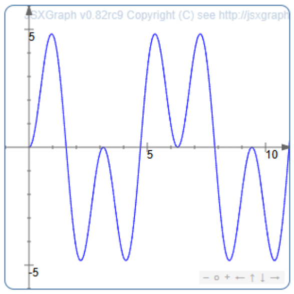
Problem 14

Match the following graphs of a function vs. time with the corresponding graph showing the location of the poles of the Laplace transform.

(1)



(2)



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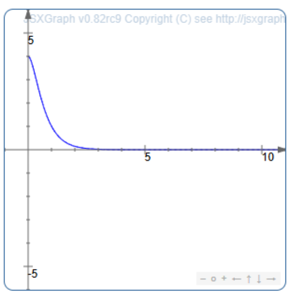

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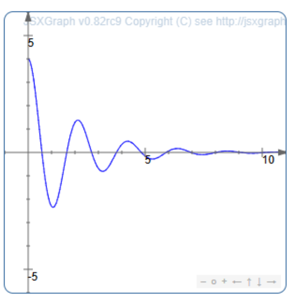

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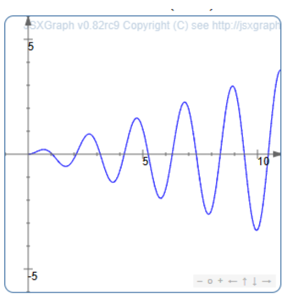
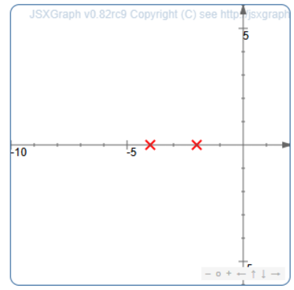
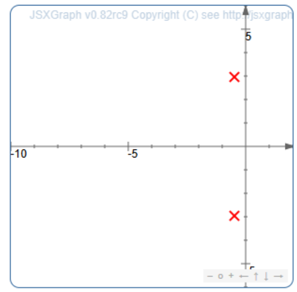


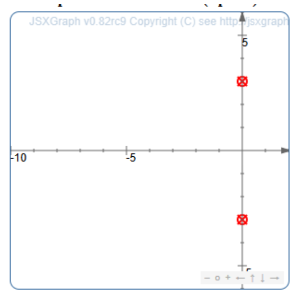
(A)



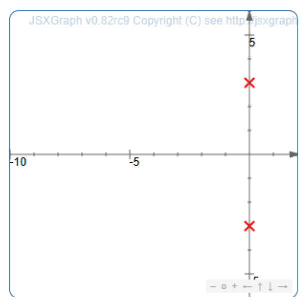
(B)



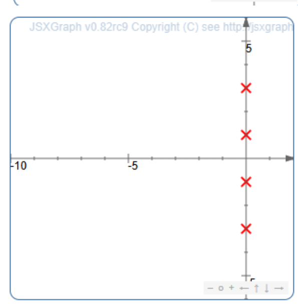
(C)



(D)



(E)



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